

# Impact of AI Adoption on Workforce Attrition

*An end-to-end data analytics project combining SQL, Power BI, and a GenAI-powered Q&A layer to quantify how AI adoption relates to employee burnout, job satisfaction, and attrition risk.*

**Tools: SQL (Google Big Query), Power BI, Python (Google Colab), Generative AI**

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## Business Problem Statement

As organizations accelerate the adoption of artificial intelligence tools across their workflows, early efficiency gains are frequently prioritized over long-term workforce impact. There is growing concern that heavier reliance on AI may be introducing unintended consequences for employees.

Rising burnout, growing fear of job displacement, and increasing attrition risk are emerging as potential side effects of AI-driven work environments. At the same time, companies are progressing through different stages of AI maturity - from early experimentation to full AI-first operating models without clear visibility into how these transitions affect employee wellbeing and retention.

Without a structured analysis, organizations risk optimizing short-term productivity gains while overlooking deeper workforce instability that could surface as higher turnover and disengagement.

## Aim and Goal of the Project

The aim of this project is to evaluate the hidden costs of AI adoption by examining how AI usage intensity, tool dependency, and organizational AI maturity influence:

- Employee burnout
- Perceived job security
- Job satisfaction
- Attrition risk

The goal is to identify patterns that signal workforce strain, over-reliance on AI, and retention risk enabling organizations to take proactive steps before these risks escalate. Specifically, the analysis tests whether rapid AI adoption is quietly increasing burnout, job insecurity, and attrition risk across the workforce, and which roles and industries are most exposed.

## Dataset Summary and Preparation

The analysis is built on an employee-level dataset of 1,500 tech-sector employees (2026), loaded into Google BigQuery, covering multiple industries, job roles, and stages of organizational AI adoption. Each record captures demographic and role information alongside AI-usage and wellbeing indicators, including:

- Role and organizational context: job role, years of experience, education level, country, industry, company size, team size, remote work type, salary
- AI usage indicators: primary AI tool, number of AI tools used per day, hours of daily AI assistance, % of tasks replaced by AI, AI adoption stage, weekly AI upskilling hours
- Wellbeing and outcome indicators: productivity score, burnout score, job satisfaction (1-5), fear of AI replacement, and attrition risk (Low / Medium / High)

Before analysis, the dataset was profiled in BigQuery: row counts were confirmed and every field was checked for missing values using a single COUNTIF audit query across all 20 columns, so that any nulls would be caught before they distorted downstream averages and rates. Once the data was validated, the analysis moved on to KPI calculation and segmentation.

## Data Analysis

### SQL Analysis (Google BigQuery)

With the data validated, five core workforce KPIs were calculated directly in SQL to establish a baseline for the whole population:

| Metric                                | Value       |
|---------------------------------------|-------------|
| Average burnout score                 | 50.06 / 100 |
| Average productivity score            | 57.54 / 100 |
| Employees at high attrition risk      | 5.67%       |
| Average share of tasks replaced by AI | 41.25%      |
| Average daily AI usage                | 4.23 hours  |

Four targeted questions were then answered with grouped SQL queries:

- Which roles are most impacted by AI in terms of task replacement?

- How does attrition risk vary across different stages of AI adoption?
- How widespread is fear of job displacement due to AI among employees?
- Which industries have the most burnt-out, dissatisfied employees at high risk of leaving?

## Power BI Dashboard

The SQL outputs were modeled into an interactive Power BI dashboard with DAX measures for average burnout, average productivity, average AI hours, task-replacement rate, and high-attrition-risk percentage, with slicers for country, industry, job role, team size, and remote work type. Key findings from the dashboard:

Across 12 job roles, engineering roles report the highest share of tasks replaced by AI (about 42–43.5%), while Product Managers, AI Researchers, and Prompt Engineers report the lowest (about 36–38.5%):

| Job role          | Avg. task-replacement % |
|-------------------|-------------------------|
| Backend Engineer  | 43.53%                  |
| Software Engineer | 43.39%                  |
| Data Analyst      | 42.58%                  |
| Cloud Architect   | 42.58%                  |
| Data Scientist    | 42.55%                  |
| DevOps Engineer   | 42.03%                  |
| AI Ethics Officer | 42.02%                  |
| ML Engineer       | 41.85%                  |
| Frontend Engineer | 41.64%                  |
| Product Manager   | 38.49%                  |
| AI Researcher     | 36.96%                  |
| Prompt Engineer   | 36.44%                  |

High attrition risk stays in a narrow band (5.08%–6.40%) across all four adoption stages, with no clear upward or downward trend as organizations move from experimentation to AI-first:

| AI adoption stage | High risk | Medium risk | Low risk |
|-------------------|-----------|-------------|----------|
| Experimenting     | 5.59%     | 43.02%      | 51.40%   |
| Integrating       | 5.08%     | 47.76%      | 47.15%   |
| Optimizing        | 6.00%     | 47.00%      | 47.00%   |
| AI-First          | 6.40%     | 45.60%      | 48.00%   |

Fear of AI replacement is widespread: of the 1,500 employees, 619 (41.27%) report medium fear, 523 (34.87%) low fear, and 358 (23.87%) high fear — so about 65% report at least a moderate level of concern about being replaced by AI.

Among employees with both high fear of AI replacement and high attrition risk, Automotive shows the highest average burnout (65.00) and the lowest job satisfaction (2.47), followed by Consulting and SaaS on burnout. EdTech shows the lowest burnout (54.29). Note: this cohort is small (85 employees are at high attrition risk overall), so industry averages within it should be read as directional.

| Industry      | Avg. years of experience | Avg. burnout score | Avg. job satisfaction (1-5) |
|---------------|--------------------------|--------------------|-----------------------------|
| Automotive    | 9.14                     | 65.00              | 2.47                        |
| Consulting    | 14.40                    | 63.00              | 2.72                        |
| SaaS          | 9.00                     | 62.67              | 2.65                        |
| Cybersecurity | 8.33                     | 60.78              | 2.81                        |
| Gaming        | 8.88                     | 60.75              | 2.60                        |
| E-commerce    | 9.85                     | 60.46              | 2.65                        |
| Media         | 12.22                    | 60.00              | 2.88                        |
| Healthtech    | 8.60                     | 57.30              | 2.74                        |
| Fintech       | 12.40                    | 56.00              | 2.88                        |
| EdTech        | 6.43                     | 54.29              | 2.83                        |

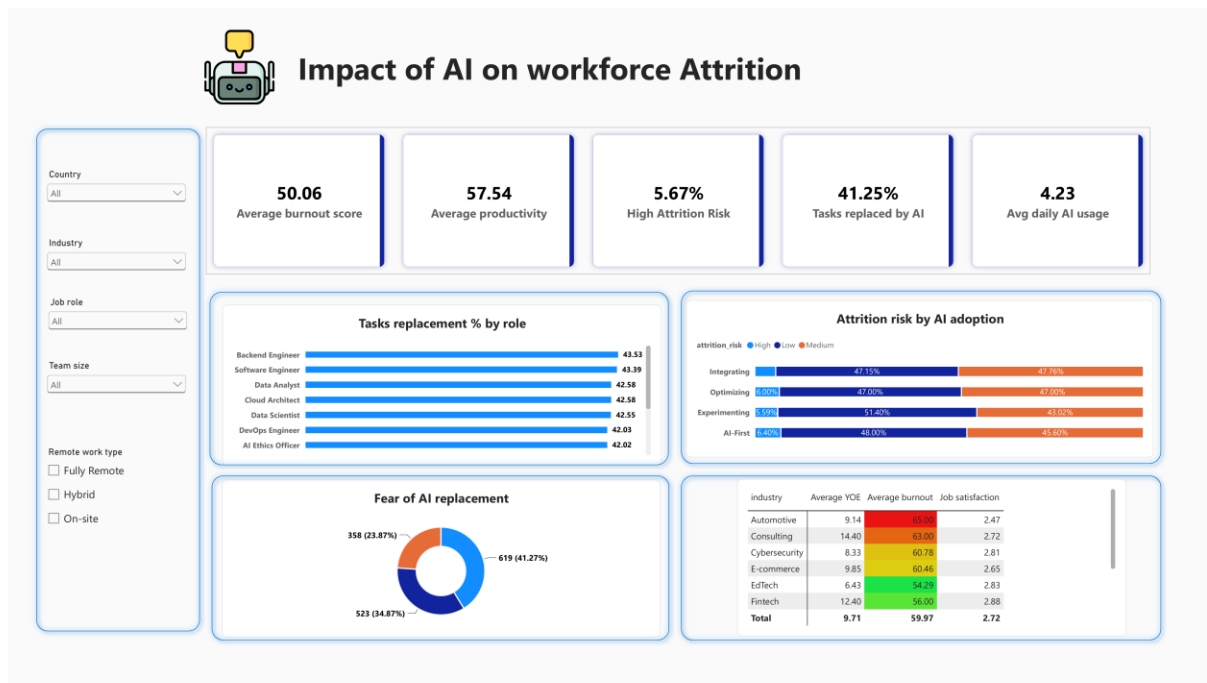


Figure 1. Power BI dashboard — Impact of AI on Workforce Attrition

## GenAI Q&A Layer (Python / Google Colab)

To make the findings accessible beyond the dashboard, the same SQL queries were run from Google Colab against BigQuery using the BigQuery Python client, and a generative-AI layer was built on top of the results:

- Query results (KPIs, task replacement by role, attrition by adoption stage, fear of replacement, and the high-risk industry cohort) were loaded into pandas DataFrames and converted to text context.
- A prompt-driven function was written for Gemini 2.5 Pro (via Google Vertex AI). It casts the model as a senior data analyst, injects the query results as context, and instructs it to answer in 3-4 structured sentences covering business impact and actionable recommendations, using only the provided data and not assuming missing values.
- The function was tested with a natural-language business question - “How can the automotive industry reduce the risk of attrition?” - and returned a grounded answer citing Automotive’s 65.00 burnout score and 2.47 job satisfaction, and recommending that AI be used to reduce workload while communicating clearly to manage fear of replacement.

- The function was then wrapped in a Streamlit app that displays the Power BI dashboard alongside a question box, and was served publicly through an ngrok tunnel, so non-technical stakeholders can ask questions in plain language without writing SQL.

## Key Insights and Recommendations

- Concern about AI replacement is widespread - about 65% of employees report medium or high fear yet only 5.67% are at high attrition risk, so perceived job insecurity is currently running well ahead of actual turnover risk.
- Attrition risk does not fall as companies mature in their AI adoption; it stays between 5.08% and 6.40% across all four stages (highest in AI-First at 6.40%), suggesting that maturity alone does not resolve workforce strain.
- Automotive stands out with the highest burnout (65.00) and lowest job satisfaction (2.47) among high-risk, high-fear employees, with Consulting and SaaS also elevated , making these priority segments for retention interventions such as workload review, transparent AI-adoption communication, and upskilling support.
- Task replacement varies by role: engineering roles see the most (about 42-43.5%) while Prompt Engineers, AI Researchers, and Product Managers see the least (about 36-38.5%), so role-level exposure to automation is not uniform.

Taken together, the analysis suggests organizations should pair AI rollout with proactive communication and support , particularly in high-burnout industries rather than assuming that reaching a more mature AI-adoption stage will, on its own, reduce attrition risk.